Bézout and integer equations

Advanced high school · extension · Worked solutions · 04

Exercise 1. Find $\gcd(84, 30)$ and a Bézout identity.

Solution. $84 = 2 \times 30 + 24$ and $30 = 24 + 6$. Hence $6 = 3 \times 30 - 84$ and the gcd is 6.

Exercise 2. Does $12x + 18y = 5$ have integer solutions?

Solution. No: $\gcd(12, 18) = 6$ does not divide 5.

Exercise 3. Find all integer solutions of $5x + 3y = 1$.

Solution. $(-1, 2)$ is a solution. All are $x = -1 + 3t, y = 2 - 5t$ for integer t.

Exercise 4. Among solutions of $5x + 3y = 31$, find those with nonnegative x and y.

Solution. Modulo 3, $x \equiv 2$. Since $0 \leq x \leq 6$, x is 2 or 5. The pairs are $(2, 7)$ and $(5, 2)$.

Problem · Investigate, justify, conclude. An order of 100 items must be delivered in boxes of 7 and 11, without opening a box. Find every possible combination, then the one using the fewest boxes. Justify that no nonnegative solution has been missed.

Solution. Solve $7x+11y = 100$. Modulo 7, $4y \equiv 2$, so $y \equiv 4 \pmod 7$. Since $0 \leq y \leq 9$, only $y = 4$ works, giving $x = 8$. This requires 12 boxes. The solution is unique, so it automatically minimises the number of boxes.